EE115 Analog Circuits
模拟电路

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Outline

- Transistor amplifier 3
- SEDTRA/SMITH book page 454-479；

Review:Drain Current vs Gate Voltage

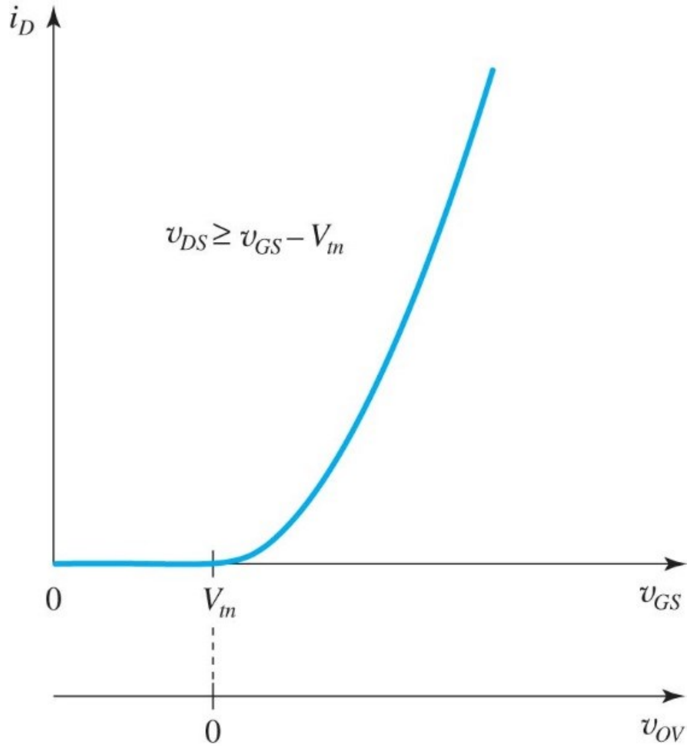


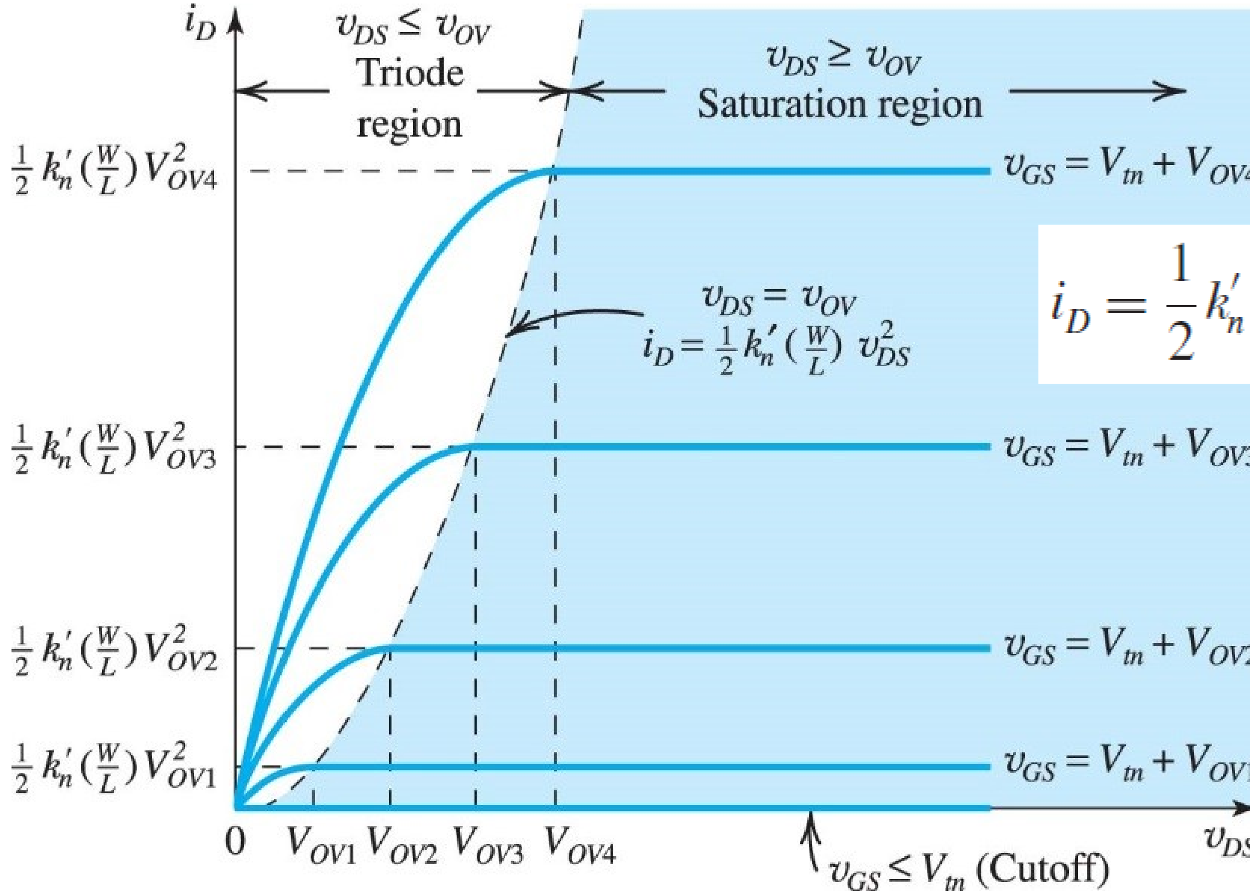
Figure 5.14 The i_D-v_{GS} characteristic of an NMOS transistor operating in the saturation region. The i_D-v_{OV} characteristic can be obtained by simply relabeling the horizontal axis, that is, shifting the origin to the point $v_{GS} = V_{tn}$.

- In Saturation Region

$$i_D = \frac{1}{2} k_n' \left(\frac{W}{L} \right) (v_{GS} - V_{tn})^2$$

Review:I-V Curves of NMOS

$$i_D = k_n'\left(\frac{W}{L}\right)\left(V_{OV}\,v_{DS} - \frac{1}{2}v_{DS}^2\right)$$



$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)v_{OV}^2$$

Figure 5.13 The i_D - v_{DS} characteristic for an enhancement-type NMOS transistor.

Review-Small signal model for MOSFET

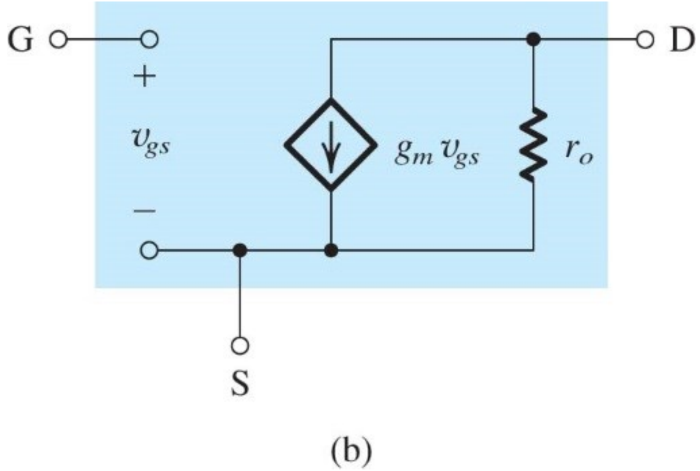


Figure 7.13 Small-signal models for the MOSFET: (b) including the effect of channel-length modulation, modeled by output resistance $r_o = |V_A|/I_D$. These models apply equally well for both NMOS and PMOS transistors.

- The equivalent circuit is valid for both NMOS and PMOS
- In PMOS, use absolute sign for all parameters: $|V_{GS}|$, $|V_t|$, $|V_{OV}|$, $|V_A|$, and replace k_n with k_p

Transconductance:

$$g_m \equiv \left.\frac{\partial i_D}{\partial v_{GS}}\right|_{v_{GS}=V_{GS}}$$

$$g_m \equiv \frac{i_d}{v_{gs}} = k_n(V_{GS} - V_t)$$

$$g_m = k_n V_{OV}$$

Vov= $\sqrt{2I_D/(k_n'(W/L))}$

$$g_m = \sqrt{2k_n'}\sqrt{W/L}\sqrt{I_D}$$

Output resistance at drain:

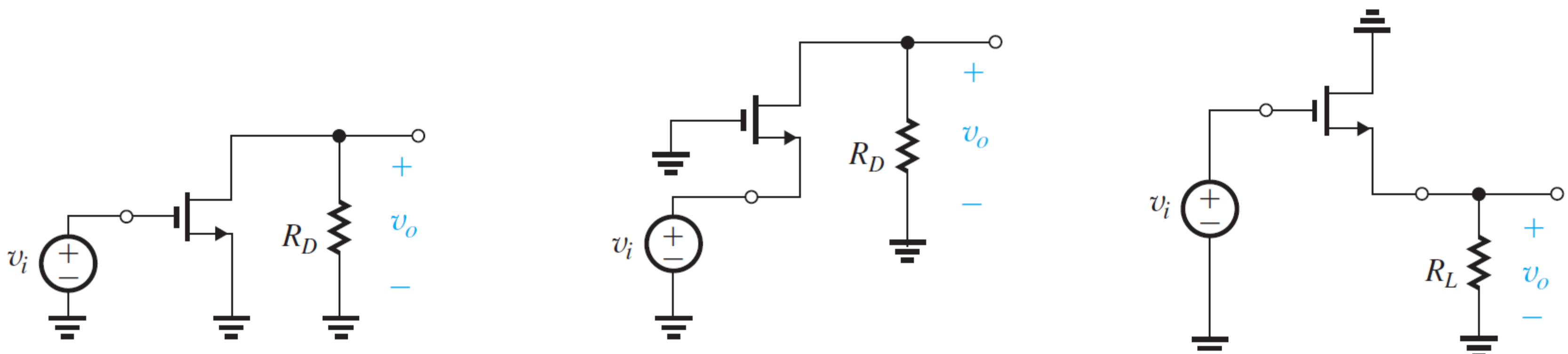
$$r_o = \frac{|V_A|}{I_D}$$

$$i_D = \frac{1}{2}k_n\left(V_{GS} + v_{gs} - V_t\right)^2$$

$$= \frac{1}{2}k_n(V_{GS} - V_t)^2 + k_n(V_{GS} - V_t)v_{gs} + \frac{1}{2}k_n v_{gs}^2$$

$$i_d = k_n(V_{GS} - V_t)v_{gs}$$

Review: Basic Single-Transistor Amplifier Configurations



(a) Common Source (CS)

(b) Common Gate (CG)

(c) Common Drain (CD) or Source Follower

Figure 7.33 The basic configurations of transistor amplifiers. (**a**)–(**c**) For the MOSFET

Review: Summary of MOSFET Amplifiers

Table 6.4 Characteristics of MOSFET Amplifiers

	Characteristics[a]				
Amplifier type	R_{in}	A_{vo}	R_o	A_v	G_v
Common source (Fig. 6.35)	∞	$-g_m R_D$	R_D	$-g_m(R_D \parallel R_L)$	$-g_m(R_D \parallel R_L)$
Common source with R_s (Fig. 6.37)	∞	$-\dfrac{g_m R_D}{1+g_m R_s}$	R_D	$\dfrac{-g_m(R_D \parallel R_L)}{1+g_m R_s}$ $-\dfrac{R_D \parallel R_L}{1/g_m + R_s}$	$-\dfrac{g_m(R_D \parallel R_L)}{1+g_m R_s}$ $-\dfrac{R_D \parallel R_L}{1/g_m + R_s}$
Common gate (Fig. 6.39)	$\dfrac{1}{g_m}$	$g_m R_D$	R_D	$g_m(R_D \parallel R_L)$	$\dfrac{R_D \parallel R_L}{R_{sig} + 1/g_m}$
Source follower (Fig. 6.42)	∞	1	$\dfrac{1}{g_m}$	$\dfrac{R_L}{R_L + 1/g_m}$	$\dfrac{R_L}{R_L + 1/g_m}$

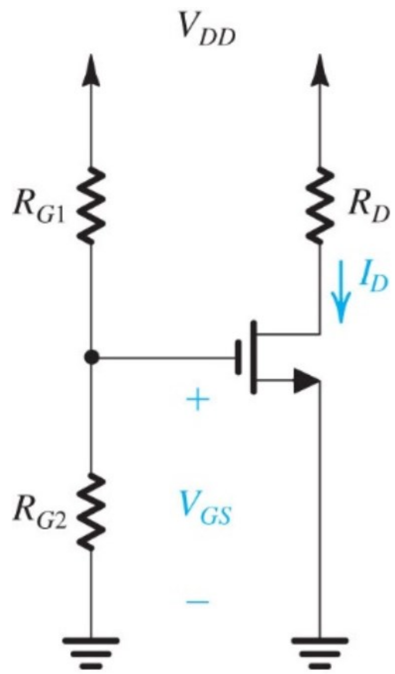
[a] For the interpretation of R_{in}, A_{vo}, and R_o, refer to Fig. 6.34(b).

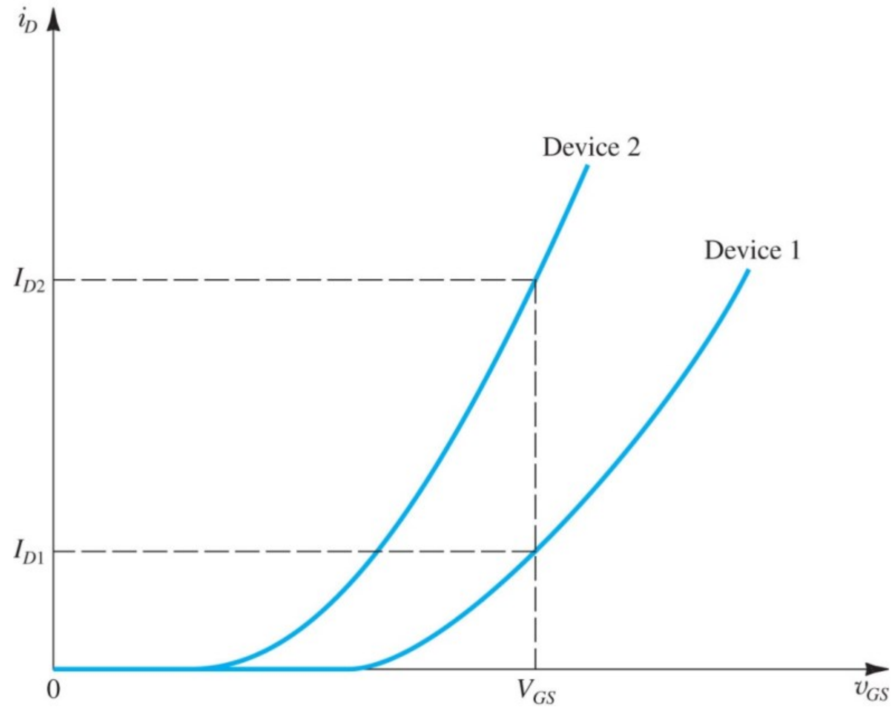
Purposes of Bias Circuit

- Provide a stable bias drain current
- Bias insensitive to variations in
  - Transistor parameters (V_t, k for MOS),
  - Temperature and other environmental changes

- Keep transistors in flat part of the I-V curves (Saturation for MOS) for desired output swing

Example of Non-Ideal Bias



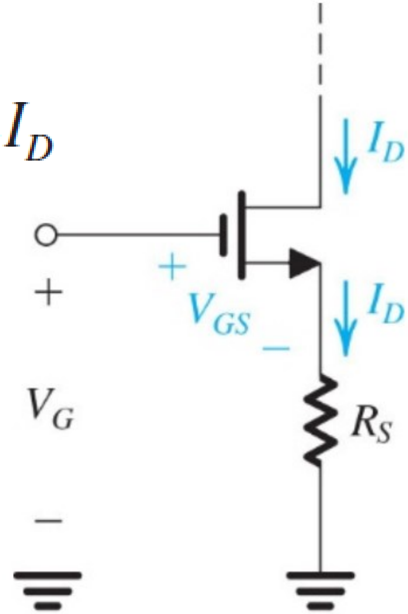


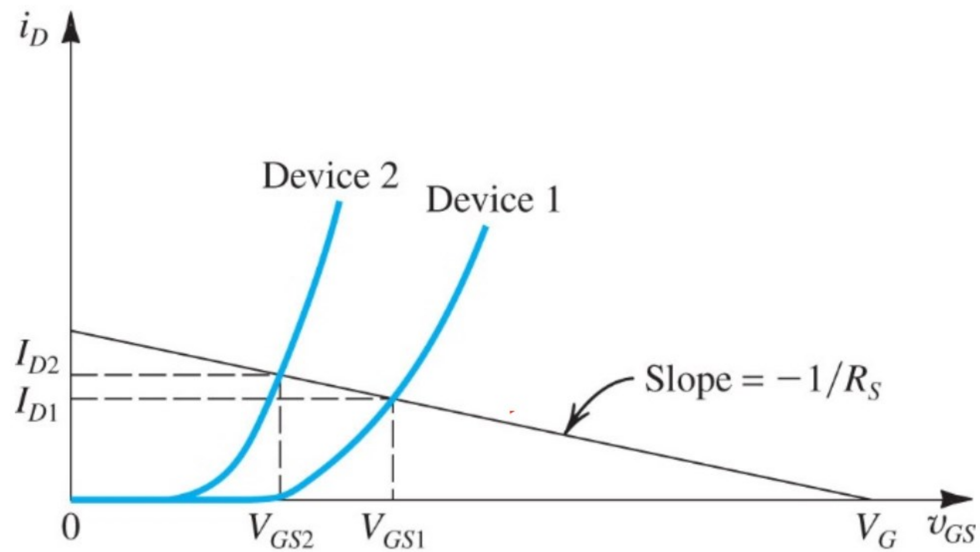
- Fixed v_{GS} from voltage divider
- Large variation in i_D due to device variations
- Large temperature dependence
  - μ_n and V_t are temperature sensitive

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_t)^2$$

Adding Source Resistance Rs

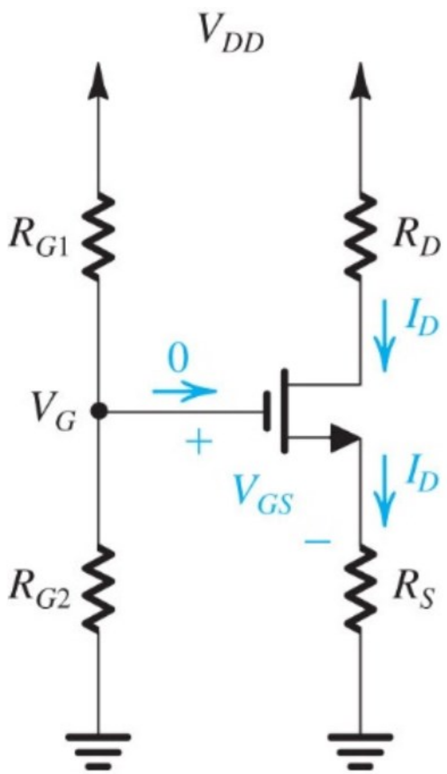
$$V_G = V_{GS} + R_S I_D$$



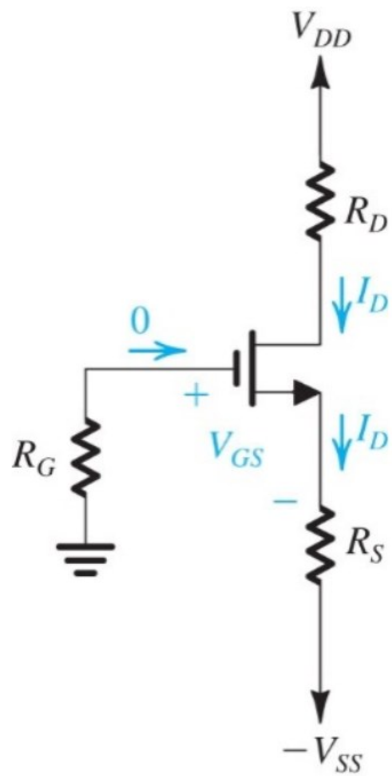


- R_S provides a negative feedback
  - If I_D increases, V_{GS} decreases -> reduce I_D
  - If I_D decreases, V_{GS} increases -> increase I_D
- Stabilize I_D w.r.t. device and temperature variations

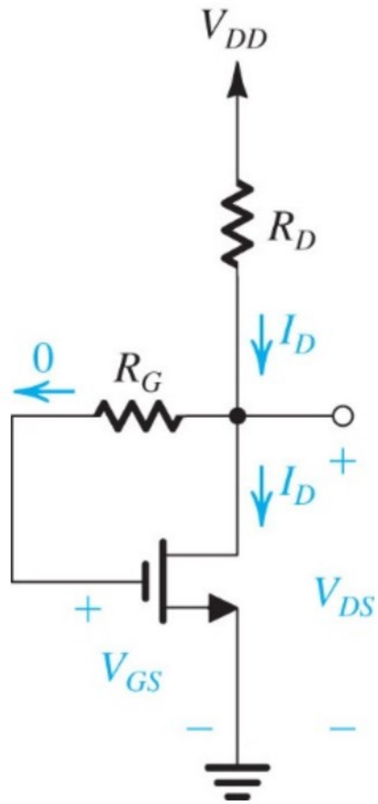
Practical Implementations



Single Power Supply



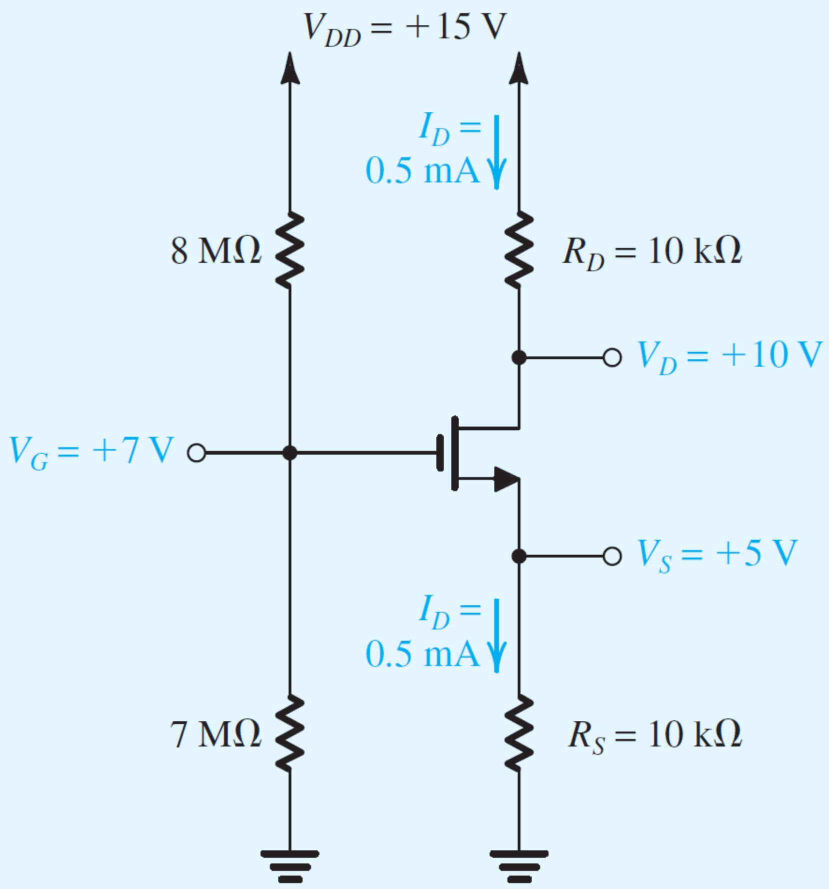
Dual Power Supply



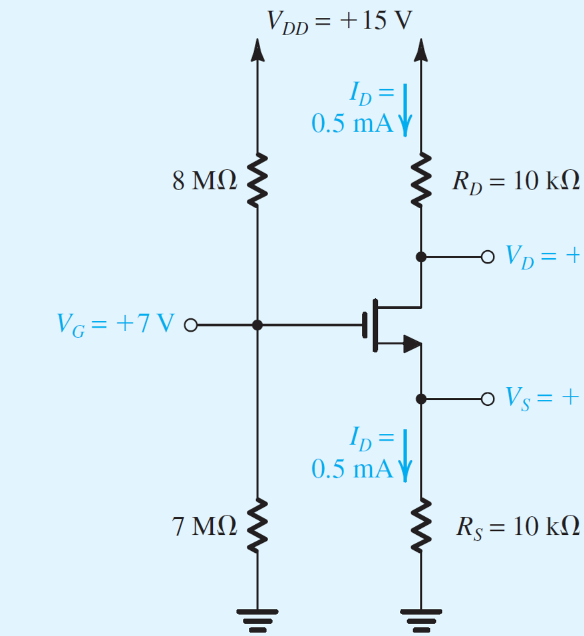
Drain-to-Gate Feedback

Example 1

It is required to design the circuit of Fig. 7.48(c) to establish a dc drain current $I_D = 0.5$ mA. The MOSFET is specified to have $V_t = 1$ V and $k_n' W/L = 1$ mA/V^2. For simplicity, neglect the channel-length modulation effect (i.e., assume $\lambda = 0$). Use a power-supply $V_{DD} = 15$ V. Calculate the percentage change in the value of I_D obtained when the MOSFET is replaced with another unit having the same $k_n' W/L$ but $V_t = 1.5$ V.



Example 1 s



Solution

As a rule of thumb for designing this classical biasing circuit, we choose R_D and R_S to provide one-third of the power-supply voltage V_{DD} as a drop across each of R_D, the transistor (i.e., V_{DS}), and R_S. For $V_{DD} = 15$ V, this choice makes $V_D = +10$ V and $V_S = +5$ V. Now, since I_D is required to be 0.5 mA, we can find the values of R_D and R_S as follows:

$$R_D = \frac{V_{DD} - V_D}{I_D} = \frac{15 - 10}{0.5} = 10\,\text{k}\Omega$$

$$R_S = \frac{V_S}{R_S} = \frac{5}{0.5} = 10\,\text{k}\Omega$$

The required value of V_{GS} can be determined by first calculating the overdrive voltage V_{OV} from

$$I_D = \frac{1}{2} k_n'(W/L)V_{OV}^2$$

$$0.5 = \frac{1}{2} \times 1 \times V_{OV}^2$$

which yields $V_{OV} = 1$ V, and thus,

$$V_{GS} = V_t + V_{OV} = 1 + 1 = 2\text{ V}$$

Now, since $V_S = +5$ V, V_G must be

$$V_G = V_S + V_{GS} = 5 + 2 = 7\text{ V}$$

To establish this voltage at the gate we may select $R_{G1} = 8$ MΩ and $R_{G2} = 7$ MΩ. The final circuit is shown in Fig. 7.49. Observe that the dc voltage at the drain (+10 V) allows for a positive signal swing of +5 V (i.e., up to V_{DD}) and a negative signal swing of 4 V [i.e., down to $(V_G - V_t)$].

If the NMOS transistor is replaced with another having $V_t = 1.5$ V, the new value of I_D can be found as follows:

$$I_D = \frac{1}{2} \times 1 \times (V_{GS} - 1.5)^2 \qquad (7.138)$$

$$V_G = V_{GS} + I_D R_S$$

$$7 = V_{GS} + 10 I_D \qquad (7.139)$$

Solving Eqs. (7.138) and (7.139) together yields

$$I_D = 0.455\text{ mA}$$

Thus the change in I_D is

$$\Delta I_D = 0.455 - 0.5 = -0.045\text{ mA}$$

which is $\frac{-0.045}{0.5} \times 100 = -9\%$ change.

Another bias circuit

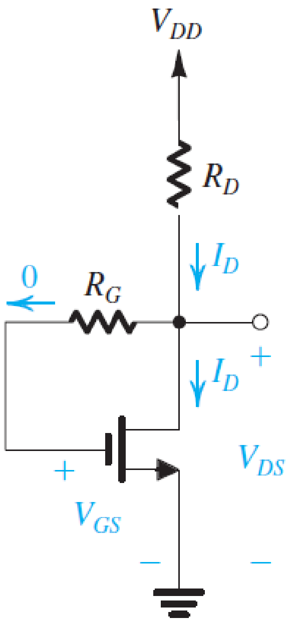
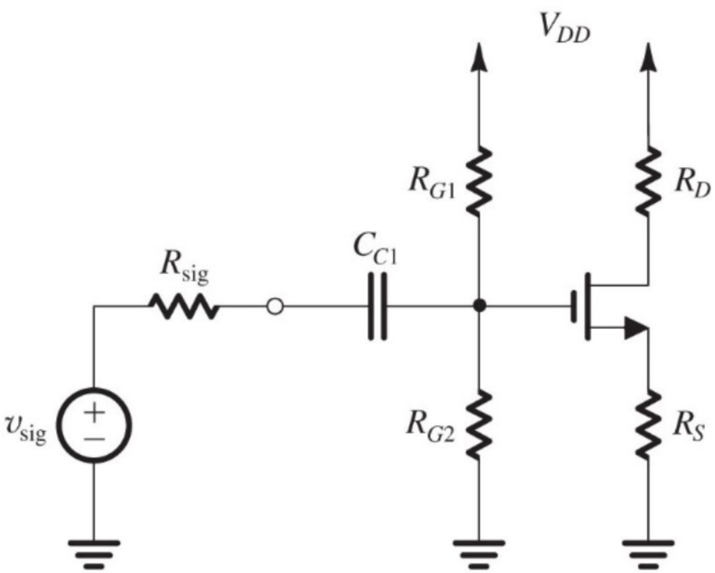


Figure 7.50 Biasing the MOSFET using a large drain-to-gate feedback resistance, R_G.

$$V_{GS} = V_{DS} = V_{DD} - R_D I_D$$

$$V_{DD} = V_{GS} + R_D I_D$$

Coupling Capacitor to Separate AC Signal from DC Bias

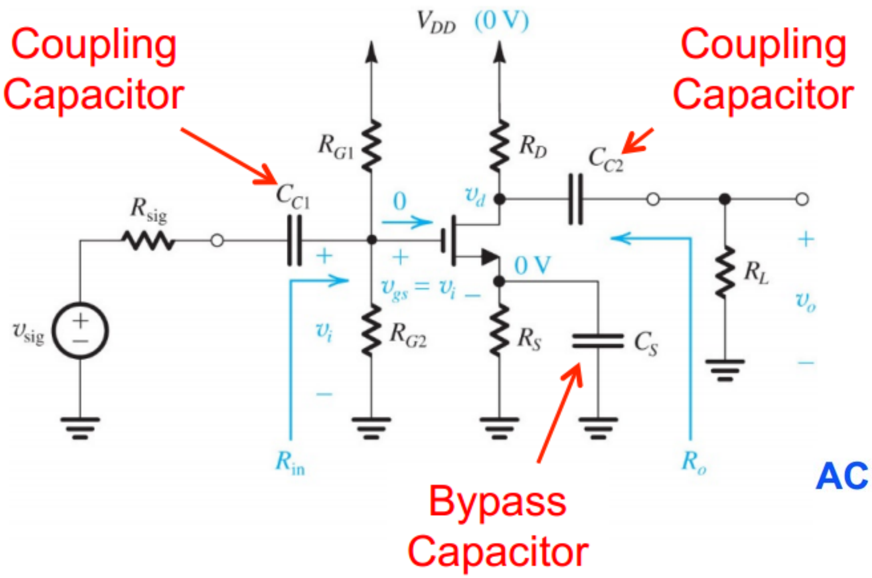


- Coupling capacitor is DC-open and AC-short
- Prevent signal source to change bias condition
- The coupling capacitor creates a lower bound of operating frequency

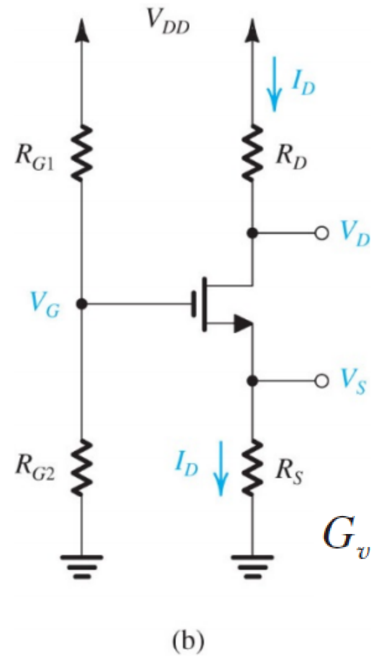
CS Amplifier with Bias Circuit

Both coupling and bypass capacitors are DC-open and AC-short

Capacitively Coupled Amplifier



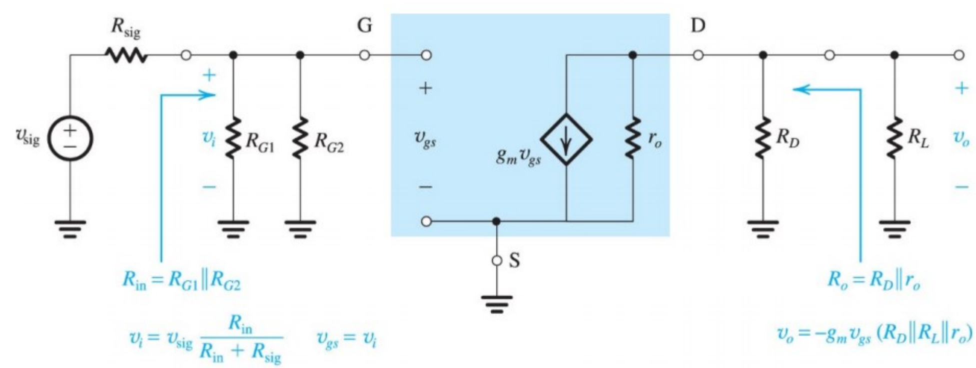
DC:



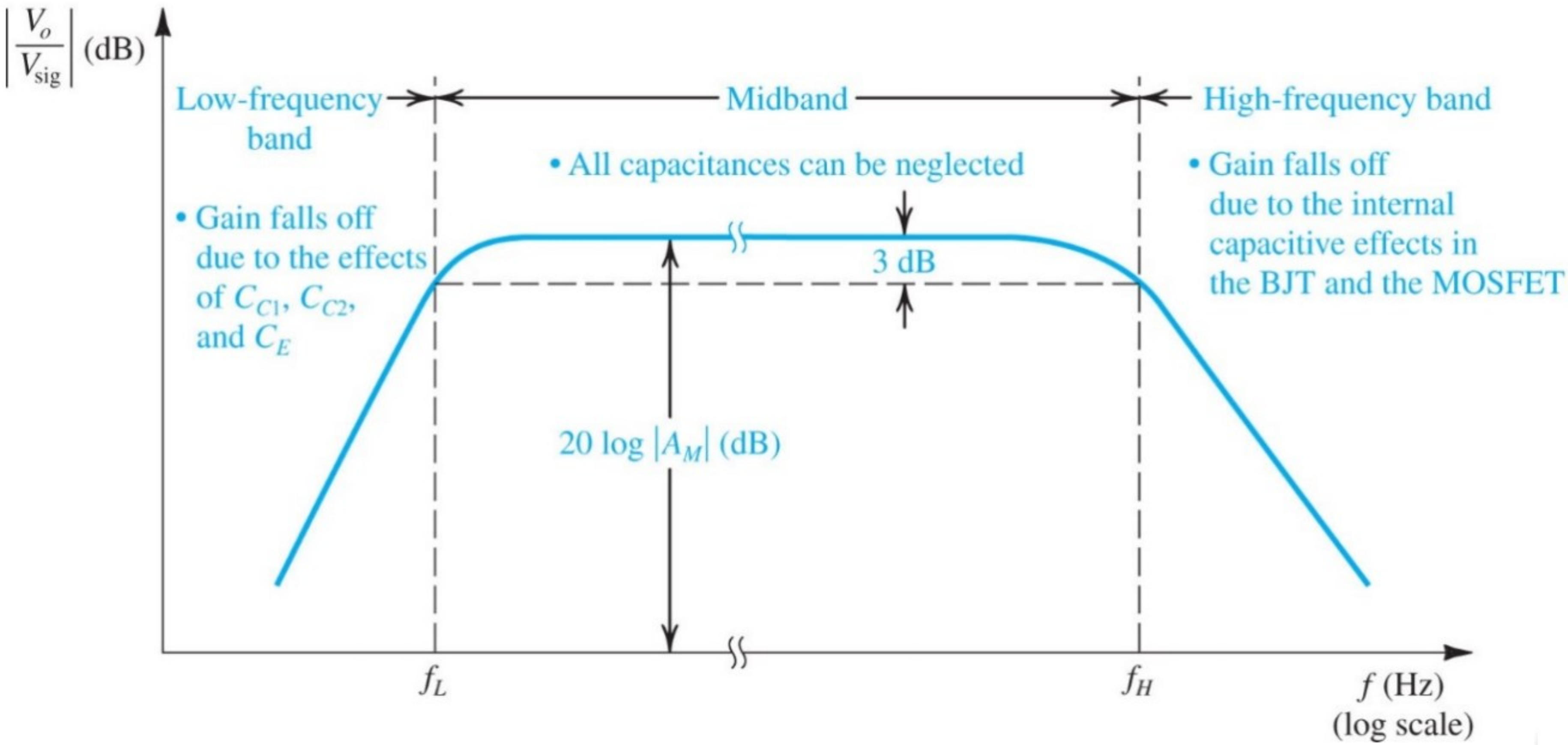
$$R_{\text{in}} = R_{G1} \| R_{G2}$$

$$G_v = -\frac{R_{\text{in}}}{R_{\text{in}} + R_{\text{sig}}} g_m (R_D \| R_L \| r_o)$$

AC:



Typical Frequency Response of Capacitively Coupled Amplifier



Homework5-due April 22nd 11p.m.

SEDRA/SMITH Book chapter 7 problem 7.92，7.98，7.102, 7.117, 7.118